

A SVM Binary Classifier for Automated Detection of Tomato Leaf Diseases in Low-cost
Greenhouse Environments

by

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Abstract

Plant diseases significantly reduce agricultural yield, putting the global food supply at risk. Traditional pathogen detection methods rely on plant pathology expertise. Moreover, traditional methods are destructive, labor-intensive, and time-consuming. Thus, plant phenotyping image processing tools are used as an alternative for detecting plant pathogens; however, these tools can be inflexible, commercially owned, and expensive. Consequently, there is demand for flexible, reliable, open-source image processing tools for plant pathogen detection in low-cost agricultural environments. Accordingly, a support vector machine (SVM) binary classifier was developed to detect tomato leaf diseases. This paper presents and evaluates the performance of this image processing tool. The classifier is trained using textural parameters calculated from gray-level co-occurrence matrices. These parameters are extracted from images of tomato leaves in the publicly available PlantVillage dataset. Overall, the image processing tool is shown to be reliable and efficient in detecting tomato leaf diseases. The underlying implementation is open-source and utilizes modular computer vision libraries, enabling other researchers to modify or extend it. Additionally, the classifier can be turned into a mobile application for low-cost automated greenhouse management. Overall, the developed image processing tool is flexible, reliable, modifiable, and efficient in detecting tomato plant leaf pathogens.

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Introduction

Plant diseases can have devastating effects on crops, leading to reduced overall yield [1, 2]. This is particularly concerning for the tomato plant, which serves as a critical global crop that many populations rely upon daily for food security and sustenance [3]. All approaches for developing disease-resistant crops or detecting plant pathogens require measurements of observable plant traits (phenotypes) [4]. Traditional plant phenotyping methods in plant pathogen detection require expertise. Moreover, these methods are destructive, labour-intensive, and time-consuming [5]. Therefore, traditional plant phenotyping methods are not suitable for monitoring large plant population sizes over time [6, 7, 8, 9]. As an alternative, non-destructive plant phenotyping methods using image processing have been developed [10], which are efficient, affordable, and allow larger samples of plants to be examined over time. However, these image-based solutions can be costly, making them inaccessible to less affluent plant growers [11, 12, 13].

Existing low-cost image-based phenotyping systems are patented, rendering the hardware and software of these systems inflexible and unmodifiable [14]. Furthermore, many image processing tools in plant phenotyping are designed for specific plant species and environments and therefore cannot be generalized to other plant species and contexts [7, 15]. Consequently, there is a need for low-cost, flexible, modifiable, generalizable, and extendable automated image processing tools for plant phenotyping in the field of plant pathogen detection.

Recent developments in image-based plant phenotyping systems research show an increased development of plant phenotyping systems that use open-source hardware [16, 13] and software libraries [6]. Despite these advances, several challenges still remain regarding data integrity and model validation. Many systems currently proposed for tomato leaf disease detection utilize popular, large-scale image datasets such as PlantVillage. However, these datasets are often imbalanced; consequently, the high performance reported in these studies is not necessarily indicative of true performance or generalizability when deployed in varying real-world scenarios.

In low-cost settings, commonly used strategies to manage the issues of large-scale image analysis and processing of phenotypic data include low-complexity traditional segmentation methods [13], the use of more affordable and straightforward imaging sensing technologies in low-cost off-the-shelf hardware [12], and image compression techniques [17]. However, each of these strategies comes with trade-offs in terms of precision, accuracy, and phenotypic analysis capabilities.

Low-complexity traditional segmentation methods struggle to handle component-based phenotyping of specific organs in plants while requiring fewer computational resources [18]. RGB sensors are affordable and generate less raw data than other methods, such as hyperspectral imaging; however, their use is limited to the phenotyping of physiological plant traits [15].

Image compression techniques can be used to reduce the amount of data transmission required for remote analysis but have a non-negligible impact on the accuracy of image segmentation [17]. Therefore, a delicate balance must be struck between cost and numerous factors that affect the overall precision, accuracy, and measurability of phenotypic traits [19].

The primary aim of this research is to create an open-source, adaptable, and extensible tool for detecting diseases in tomato plant leaves, specifically tailored for low-cost automated greenhouse settings. Additionally, this study strives to design a solution capable of generalizing to data beyond the original training dataset while comparing its performance against existing related systems. By achieving these objectives, this research will deliver an economical, efficient, accessible, and flexible system for low-cost tomato leaf disease detection.

To address the challenges associated with image-based phenotyping solutions for plant pathogen detection, and the above-mentioned goals, this paper proposes the development of a binary support vector machine (SVM) classifier [20] for detecting tomato plant leaf diseases. SVMs were chosen due to their proven effectiveness in numerous successful implementations of tomato leaf disease detection [3, 21, 22]. Additionally, the availability of open-source software libraries for SVM development aligns with our goal of creating a flexible, adaptable, and accessible tool. By leveraging these resources, we can ensure that our solution is both practical and comparable to existing systems. The extensive application of SVMs also provides a foundation for comparing our results, enabling us to evaluate whether our approach and existing solutions generalize effectively across varying condition, thereby determining their true efficacy. The proposed classifier is trained using the publicly available PlantVillage dataset [23], and its overall reliability and performance are assessed.

The remainder of this paper is organized as follows: Section 2, the literature review, presents the fundamentals of plant phenotyping and its relation to plant pathogen detection. Thereafter, in section 3, the methodology, we discuss the

proposed classification framework, detailing how images are processed, how the SVM classifier is trained, and how the overall system’s performance and reliability are evaluated. In section 4, we discuss our experimental results after our SVM classifier has been trained and evaluated. Lastly, in section 5, we discuss possible extensions of our classifier, potential future research directions, and implications of our experimental results.

Literature Review

2.1 The problem of plant diseases

Our lives would not be possible without agriculture. We use goods like medicine, pesticides, and rubber everyday, all of which are made from plants. Plants are ubiquitous in our lives, and they fulfill some of our essential needs. Most importantly, plants feed most people globally [24]. But many of us still suffer from hunger. In 2024, it was estimated that between 638 and 720 million people did not have enough to eat [25]. Plant diseases make this situation even worse by robbing us of food [26, 27, 28, 23]. Therefore, they are a direct threat to our livelihoods.

Plant diseases make it difficult for crops to survive and stay healthy [29]. They interfere with a plant's cells causing them to malfunction or die [30]. And each cell helps the plant carry out essential life-sustaining functions: conducting photosynthesis, and absorbing water and nutrients. So, when these cells are damaged, the plant's ability to obtain and use nutrients is limited. As a result, plant diseases stunt the growth, and decrease the yield and nutritional value of crops [31, 32, 33, 34, 35]. In some cases, the damage is so severe that farmers lose up to 100% of their crops [28, 36]. However, the losses caused by plant diseases goes further than just the yield and quality of crops; they extend to rural communities, consumers, the environment, and governments [31]. Therefore, plant diseases threaten the health of crops, and this has disastrous far-reaching effects on our livelihoods. Clearly, we need solutions to deal with them.

2.2 Some solutions to plant disease

Luckily, several options exist. For example, plant breeding is an environmentally sustainable and cost-effective solution, which helps to produce disease-resistant varieties of plants [37, 38, 27, 10]. Alternatively, we also resort to precision agriculture, which uses data from crops and their environment to make optimal decisions about agricultural actions [39, 40, 41, 42]. For instance, precision

agriculture aids farmers in deciding on the best way to apply pesticides during a pathogen epidemic [43] by helping determine when to apply pesticides, how much pesticide to apply, and where to apply it; as a result, such precise pesticide application will reduce costs, save resources, and benefit the environment and human health [44, 45, 46, 47, 48, 49]. So, plant breeding and precision agriculture help deal with plant diseases; however, these methods depend on effective plant phenotyping [6, 50, 4, 51, 52, 37].

2.2.1 The relationship between plant diseases and plant phenotyping

Plant phenotyping is the evaluation of observable or measurable plant phenotypes [53, 54], which are proxy measurements or traits indicating a plant’s stress, health or growth in relation to its environment [53, 15]. Moreover, phenotypes represent traits of an individual plant in terms of all of its chemical, physiological, morphological, and behavioral aspects [55, 56, 15]. So, plant phenotyping provides quantitative data that represents various aspects of a plant. When a plant is diseased its form, function and integrity are affected [29], therefore reliable measurements of all aspects of the plant are required in plant disease solutions. Consequently, plant phenotyping is an effective quantitative method for plant breeding and precision agriculture [4, 57, 36, 58, 37].

2.2.1.1 Plant phenotyping and plant breeding

The goal of plant breeding is to develop plants with desirable traits like disease resistance. In this regard, the most important tasks for breeders are the selection of suitable individual plants and the development of genetic material [37]. Plant phenotyping assists in these tasks. It does so by providing breeders data about a plant’s phenotype-environment interactions, which allows them to identify and select suitable parameters for the development of improved crops [59]. And this method has been used successfully in several breeding programmes for disease resistance. To name a few, [60] conducted a study for Barley Rust Stripe resistance, [61] for Oat Crown Rust resistance, [62] for Wheat Head Blight resistance, [63] for Apple Scab resistance, and [64] for Cotton Verticillium Wilt resistance. Thus, plant phenotyping effectively assists plant breeding programmes.

2.2.1.2 Plant phenotyping and precision agriculture

Precision agriculture provides data-driven decision making for optimal agricultural management, and plant phenotyping can enable this [65, 47, 45, 48, 66]. Notably, there are many pesticide, disease detection and warning systems that rely on phenotyping [46, 34]. For example, [67] discusses a warning system for controlling infections of Grapevine Downy Mildew, and [68] presents an early disease detection system for selective pesticide application which successfully helped managed diseases in apple, olive and almond farms. Evidently, precision

agriculture with phenotyping provides effective plant disease management. But traditional phenotyping has problems.

2.2.2 Traditional plant phenotyping solutions, and their challenges

Human naked eye observation is the main traditional phenotyping method used in plant disease detection. This method needs continuous effort from human raters which can be expensive. And in poorer countries, raters are far and few between [69, 70, 71], so farmers will need to expend more time, energy and resources to obtain raters [72, 73, 74]. Thus the consultation of human experts is expensive, time-consuming and not accessible to everyone. Additionally, naked eye observation has practical limitations. For example, when plant samples are needed, visual assessment is destructive, and this prevents the observation of plant traits over time [4, 74, 75]. Moreover, visual assessment is error prone when a disease does not induce any visible symptoms in a plant, thus the detection of early disease onset is difficult for human raters. However, alternative laboratory methods for early disease detection exist [7], but they can also be computationally expensive and time consuming just like human visual assessment [70, 69, 43, 40]. Fortunately, image sensing technologies combined with machine learning and image processing techniques can overcome many problems that traditional phenotyping methods suffer from.

2.2.3 Image sensing technologies an alternative to traditional phenotyping methods

Image sensing technologies overcome some of the deficiencies of traditional plant phenotyping methods. For example, traditional methods can be destructive, which complicates the measurement of plant traits over time [4, 74, 75]. And measuring phenotypes over time is important because plant disease symptoms can be latent or appear suddenly, and then progress rapidly [58, 76]. Thus, non-destructive phenotyping methods are beneficial for early disease detection and treatment. Evidently, traditional methods are lacking in this regard, but image sensing technologies are not, since they can precisely and non-destructively sample large populations [77, 27]. Moreover, this can be done in a very short period of time [78]. Therefore, image sensing technology is more economical than time-consuming human labour.

Sensors also capture plant tissue–light interactions and this enables the measurement and quantification of disease symptoms that are not visible to the eye [15]. Specifically, plant tissues absorb, reflect, emit, transmit, and fluoresce different wavelengths of light [79, 80]. Light is captured by a sensor, and this light can have different wavelengths each of which may correspond to distinct biochemical or physiological aspects of a plant [81, 82]. Therefore, sensing technologies increase the possible types of traits we can observe, which allows for

earlier, more accurate, and comprehensive disease symptom detection.

Numerous studies have demonstrated the effectiveness of sensing technology in plant disease detection and management. [83] used chlorophyll fluorescence imaging to detect leaf rust and assess its severity in wheat. [84] used thermography and chlorophyll fluorescence imaging to non-destructively visualize *Cercospora beticola* infections. [85] applies a near-infrared hyper-spectral image to detect blast rice in Nipponbare seedlings. [86] used hyperspectral imaging to detect early blight and late blight diseases in tomato leaves. [87] uses thermal image sensing to detect tea diseases. Clearly, image sensors have proven effective in supporting plant disease detection and management. Furthermore, in recent years, a wide range of machine learning and image processing techniques have been used with these sensors.

2.3 Image processing pipelines

In spite of varying characterizations of an archetypal image processing pipeline, most existing solutions in the literatures exhibit common element or phases in which machine learning and various other techniques are applied to assist in the detection of plant diseases [34, 46, 88, 89, 70, 57, 90, 91, 92, 72, 93]. These phases are, namely, image acquisition through sensors which we have discussed earlier, image preprocessing, image segmentation, feature extraction, and classification. [34, 46, 88, 89, 70, 57]

2.3.1 Preprocessing

Image preprocessing involves noise removal, image enhancement, and colour space conversion. During image acquisition, noise originating from the imaging sensor, the environment, or other external sources can compromise the quality of the data. This is problematic because such artifacts can ultimately reduce the accuracy of downstream image processing and classification phases. Therefore, preprocessing is crucial for accurate and effective plant disease detection.

Filtering is a common technique used in preprocessing to remove salt-and-pepper noise while preserving sharp edges or disease spots [94, 95, 96]. Common examples in the literature include median filtering [97, 93], Gaussian filtering [98], and bilateral filtering [99]. In addition to noise removal, image enhancement techniques like histogram equalization are used to improve the visibility of disease spots that might otherwise be obscured by poor lighting. Specifically, this method improves the contrast between healthy and diseased areas of an image [100, 101, 102].

Another technique is colour space conversion which is necessary to isolate relevant plant traits. Typically, images are captured in RGB, but many studies convert them to HSI (Hue, Saturation, Intensity) or the Lab* color space. This

conversion is preferred because it is easier to identify yellow or diseased spots when color information is separated from luminance [103, 104, 105, 5, 90, 106]. Consequently, by utilizing these preprocessing techniques, researchers can ensure that subsequent image processing phases, such as image segmentation, are performed on high-quality, reliable data. And this leads to more effective and reliable results.

2.3.2 Segmentation

Segmentation is a key image-processing task common to many image-based plant phenotyping solutions. It is the process of separating the object of interest from its background or other parts of the image which are not relevant. Often statistical metrics, and features are only desired for the object of interest. Hence including image data unrelated to the object of interest affects the accuracy and reliability of any results obtained from downstream processes in the pipeline. Therefore, the validity and precision of the derived phenotypical measurements and analysis depend on the image segmentation method selected [18, 19]. Therefore, image segmentation methods are integral to image processing in plant phenotyping.

Selecting the appropriate segmentation technique is crucial when considering the type of phenotypic parameter to be measured. Segmentation methods are typically classifiable as traditional or learning-based [18]. Traditional segmentation methods typically target holistic phenotypical traits like plant height and include frame differencing [107], colour-based [97], edge detection [108], spectral band difference-based [109], shape modelling-based [110], and graph-based segmentation [111]. In contrast to traditional techniques, learning-based methods, which utilise machine learning or deep learning [112, 113], are more suitable for targeting component-based traits of specific organs or parts of a plant [18].

2.3.3 Feature Extraction And Classification

Feature extraction gathers information from targeted regions of plant images, allowing for the analysis or classification of this data to answer questions or guide decisions about the area of interest. In plant disease detection, we extract features from plant images to quantify actual phenotypic traits, and this helps to determine if a tomato plant leaf shows signs of disease or not.

2.3.3.1 A Taxonomy of Phenotypical Traits

A phenotypical trait is classifiable as morphological, physiological, or temporal. Additionally, the literature indicates morphological and physiological phenotypical traits are classifiable as holistic-based or component-based [78]. Holistic properties consider the plant as a whole, whereas component-based features examine individual parts of the organism, such as its fruit, flower, leaves, stem, etc.

2.3.3.2 Leaf-Derived Phenotypical Traits: Robust Proxy Indicators of Plant Metabolic and Physiological Status

Leaf-derived traits are among the most used indicators of plant status because leaves rapidly respond to environmental stimuli and robustly indicate various factors that determine plant health, growth, and stress [8]. Traits of leaves can monitor and predict a host of plant health, growth and stress factors. For example, leaf area predicts plant productivity because reduced leaf area results in decreased chlorophyll content and less photosynthesis. Therefore, leaf area is a good proxy measure for evaluating plant growth and has been used to monitor and estimate crop yield and health [114].

Various other leaf morphological traits exist in plant phenotyping. These include number of leaves and inclination angle for monitoring plant growth and development [115, 8], and leaf thickness for measuring abiotic stressors like water scarcity [116]. Additionally, physiological traits of leaves exist that indicate various factors of plant status, such as stomatal conductance for detecting biotic stressors like pathogens [117] and leaf water content for drought stress [118]. Considering the above, leaf-derived phenotypical traits are robust proxies of various metabolic and physiological plant parameters.

Although leaves play an important role in evaluating plant growth, health and resilience to stress, these evaluations cannot be scaled to the whole plant [8]. Rather, other parts of the plant should be phenotyped to understand the processes governing plant status comprehensively. The literature is rife with such examples that study traits of roots [119], stems [120], seeds [121], and fruit [122].

2.4 Current State of Image-based Tomato Plant Leaf Phenotyping

In the literature there is indication of two approaches for plant leaf image disease detection [123]. The first approach is feature-based which requires the development of hand-crafted features such as color and textural features to train a traditional machine learning algorithm. And the second is deep-learning based in which features are automatically learnt.

2.4.1 Feature-based Disease Detection

There are many different types of features used in the literature and include textural and color-based features. For example, there are studies which calculate textural statistics like the Gray level co-occurrence matrices, and Hu moments which are then used to detect plant diseases [124, 125, 126, 123, 22]. In terms of color features some common types of features are color histograms, color moments, and correlograms [127, 128, 125].

Once features are determined a suitable classifier is constructed, and typically this is a traditional machine learning algorithm. There are many different types of classifiers used in the literature in leaf disease detection. For example, [5] uses Fuzzy sets and achieved an accuracy of 97.1%. [22] employed a Support Vector Machine classifier with textural features from Gray level co-occurrence matrices to detect tomato leaf diseases and achieved an accuracy of 99%. [70] used a Naive bayes classifier and achieved an accuracy of 86%. [129] used the method of Random Forest and achieved an accuracy of 95%. [130] used K Nearest Neighbours classifier to detect tomato leaf diseases and achieved 93% accuracy. [131] used a decision tree to detect tomato plant diseases and achieved 78% accuracy. Evidently, feature-based plant leaf disease detection is effective. However many studies use data sets like PlantVillage to train and validate their model, and this data set is imbalanced with images taken in uniform controlled conditions. Therefore, further investigation into the generalizability of these models is required to determine their efficacy on unseen data and in field scenarios.

2.4.2 Deep learning Disease Detection

In terms of deep learning approaches there are numerous different architectures, and solutions used. Notably there is an extensive use of Convolutional Neural Networks (CNNs) with many different types of architecture used, and accuracy rates as high as 98% [132, 133, 134, 135, 136, 137]. Despite the widespread use of CNNs and their success in plant tomato plant leaf disease detection, many studies train models on data sets which are imbalanced and with images taken in uniform lighting conditions. Consequently, this means that proposed models may not generalize well to unseen data or practical applications in the field.

2.5 Miscellaneous Issues Of Image-Based Phenotyping Systems

2.5.1 Low-cost Automated Solutions In Greenhouses

Many automated phenotypical image processing systems in greenhouses employ various advanced technologies, such as robotics [138], automated conveyor belts, and benchtop configurations [139, 140], as well as advanced image sensors in high-throughput phenotyping systems [141]. However, these solutions are not accessible to all plant growers because they tend to be costly [11, 12, 13]. Moreover, many systems are patented by specialized companies, rendering the hardware and software inflexible and unmodifiable [14]. Furthermore, existing low-cost phenotyping systems in greenhouses have bespoke image analysis and processing pipelines, which prevent them from being generalized and used in broader contexts [7, 15]. Therefore, there is a need for low-cost, flexible, modifiable, generalizable, and extendable automated image processing systems for plant phenotyping applications.

2.5.2 Open-source Software, Systems and Affordable Hardware As Cost Reducers

Recent studies in low-cost automated image-based plant phenotyping show increased development of open-source phenotyping systems and the use of readily accessible commercial off-the-shelf hardware [16, 13] to reduce the costs of image-based phenotyping in greenhouse environments. Cheap low-cost compute sensors called Raspberry PIs have been used with affordable RGB and hyperspectral sensors to monitor phenotypic traits in greenhouse environments [12]. Additionally, there has been an increase in open-source image-processing plant phenotyping systems like Phenobox [14]. Despite these advances, several challenges remain in low-cost plant phenotyping systems.

2.5.3 Challenges In Automated Image-Based Plant Phenotyping

2.5.3.1 Large Image Sensor Data Throughput Necessitates Robust Image Processing Tools

Modern image sensors in image-based phenotyping generate massive amounts of data [142], necessitating the development of efficient image processing pipelines in these contexts to handle large datasets in the shortest possible time while also delivering precise and accurate phenotypical analysis [13]. Therefore, effective choices in image sensors are crucial for developing low-cost, automated image-based phenotyping systems [8].

2.5.3.2 The Effects of Image Sensor Choice On Phenotypical Analysis And Processing

The choice of sensor has a non-trivial influence on the image analysis and processing techniques used in plant phenotyping and additionally also determines the breadth of traits that can be analysed [15]. For example, hyperspectral imaging is practical for studying plant metabolic status; however, the sensors required generate Terabytes of data when monitoring multiple plants [143]. Consequently, in low-cost settings, hyperspectral imaging of component-based phenotypical traits becomes infeasible because learning-based image techniques, which are most suitable for this purpose, are costly, time-consuming, and impractical when large amounts of phenotypical data are processed [143]. Additionally, low-cost hardware has constrained bandwidth, storage, and compute power. Therefore, remote cloud image processing and analysis are required [144], which can be expensive [13].

2.5.3.3 Solutions, and Limitations For Low-cost Large Data Throughput Image Processing Systems

In low-cost settings, commonly used strategies to manage the issues of large-scale image analysis and processing of phenotypic data include low-complexity

traditional segmentation methods [13], the use of more affordable and straightforward imaging sensing technologies in low-cost off-the-shelf hardware [12], and image compression techniques [17]. However, each of these strategies comes with trade-offs in terms of precision, accuracy, and phenotypic analysis capabilities. Low-complexity traditional segmentation methods struggle to handle component-based phenotyping of specific organs in plants while requiring fewer computational resources [18]. RGB sensors are affordable and generate less raw data than other methods, such as hyperspectral imaging; however, their use is limited to the phenotyping of physiological plant traits [15]. Image compression techniques can be used to reduce the amount of data transmission required for remote analysis but have a non-negligible impact on the accuracy of image segmentation [17]. Therefore, a delicate balance must be struck between cost and numerous factors that affect the overall precision, accuracy, and measurability of phenotypic traits [19].

2.6 Potential Advances For Low-Cost Automated Image-Based Plant Phenotyping

Numerous opportunities exist to improve the overall affordability and generalisability of plant phenotyping in automated greenhouse environments. Low-cost plant phenotyping is not a nascent field of inquiry. However, many tracts of innovation have not been fully exploited. For instance, learning-based imaging techniques for analysis and processing can reduce analysis and segmentation costs but struggle when executed on resource-constrained hardware [145]. Therefore, there is a need for the development of efficient and optimised deep-learning or machine-learning models in resource-constrained environments. Similarly, many existing phenotyping systems are not generalizable to other plant species [15, 146] and rely on proprietary software or hardware [14]. Hence, there are opportunities to develop modifiable and flexible image-processing pipelines that are amenable to a broader variety of plant phenotyping contexts. Additionally, there are not many mobile applications for plant phenotyping. Mobile devices have advanced image sensors and can locally run image analysis and processing tasks, eliminating the need for cloud computing resources that increase phenotyping costs [19].

2.7 Conclusion

Ensuring an adequate food supply in response to the growing human population is crucial [26]. Within this context, the tomato represents a globally critical crop that a significant portion of the population relies upon for daily sustenance. Image-based plant phenotyping has proven to be an effective tool for this purpose, aiding in the development of cultivars with maximum yield and resistance to various environmental stressors [37]. These methods are non-destructive, more time-efficient, replicable, and affordable than traditional methods [78]. However, image sensors in plant phenotyping generate large amounts of data, which necessitate robust image analysis and processing tools to handle this data effectively [142]. Large plant data throughput makes image-based plant phenotyping in automated greenhouse environments cost-prohibitive, as affordable off-the-shelf hardware and sensors often lack adequate local resources to store or process large amounts of data [143].

Consequently, remote cloud computing solutions handle image processing and analysis tasks [144], which can further increase operational costs in greenhouse management [13]. Image-data compression techniques, low-complexity image analysis and processing techniques, and low-cost off-the-shelf hardware are effective in reducing the cost of image-based plant phenotyping by reducing the computational requirements and data throughput [17, 12, 13]. However, these solutions come with marked trade-offs in terms of accuracy, precision, depth, and breadth of plant trait analysis. Moreover, existing solutions are often not generalizable to other environments and plant species [7, 15]. Moreover, models trained on imbalanced datasets may demonstrate high performance- as the literature indicates-but they have not been tested extensively on unseen data to determine if they generalize to varying sources of data. Therefore, their efficacy in real world scenarios has not been verified.

In short, there is a need for low-cost, generalizable, open-source, flexible, modifiable, and robust plant phenotyping image processing and analysis tools that are effective and efficient on resource-constrained hardware [13]. Future research should develop low-cost image processing tools that are precise, accurate, and adaptable to various plant monitoring contexts or parameters. In doing so, automated image-based techniques for tomato plant disease monitoring will become more accessible in low-cost greenhouse environments.

Materials and Methods

3.1 Dataset

The PlantVillage dataset is used in our study. PlantVillage is a collection of 54309 annotated diseased and healthy leaf images spanning several plant species, including tomato [23]. The size of the subset of tomato leaf images is 18162. Several images of each tomato leaf were captured outside under natural light in various lighting conditions and orientations, against a black or grey background, using a standard digital camera [23, 1]. The images are captured in the RGB colour space and have dimensions 256x256. We have selected PlantVillage as our primary dataset because it is widely used in image processing research for plant disease detection [147, 148, 149, 150, 151, 152]. Using this dataset will allow future researchers to compare our proposed method to previous studies.

3.2 Image processing pipeline

Many studies in image-based plant disease detection typically have predetermined phases in their image processing pipeline. Namely, preprocessing, segmentation, and feature extraction [149, 91, 92, 72, 93]. We follow suit and include each of those phases in our image processing pipeline.

3.2.1 Preprocessing Phase

Preprocessing involves tasks such as noise removal, image enhancement, and colour space conversion. In image acquisition, noise from the imaging sensor, the environment, or another source can compromise the accuracy of downstream image processing and classification phases. For example, we choose a global thresholding segmentation method, which we discuss later, and these methods often perform poorly in the presence of noise or uneven illumination [94]. Nevertheless, preprocessing steps such as histogram equalisation and median filtering effectively mitigate noise and illumination issues [94, 95, 96]. Thus, preprocessing is a crucial phase of image processing pipelines, which enhances the overall accuracy of segmentation and image classification [1, 153]. Therefore, we apply

several preprocessing steps to enhance the overall effectiveness and accuracy of our segmentation and classification phases.

We aim to compute texture features of the tomato leaf in each image using the grey-level co-occurrence matrix [124], which requires a grey-level image to calculate. Therefore, the first preprocessing step we perform is the conversion from the leaf image’s original RGB colour space to the grayscale colour space.

Once we obtain the grayscale image of the tomato leaf, we apply histogram equalisation. This preprocessing method is used in the literature to add more contrast between the background and foreground of an image [95]. Image contrast enhancement is achieved by uniformly distributing the grey-level intensity values of the image [94], which makes histogram equalisation particularly useful for dealing with various illumination conditions [1].

Lastly, we apply median filtering to remove common salt-and-pepper noise from the image. Median filtering is an effective image enhancement preprocessing step that has been successfully used in plant leaf disease detection [97, 93]. Noise manifests as sharp transitions in image intensity seen in the grayscale image histogram [94], therefore noise can negatively affect the accuracy of global threshold segmentation methods, which we use in this study; therefore, to enhance segmentation accuracy and robustness, we apply median filtering to denoise the image.

3.2.2 Segmentation Phase

We use the Otsu method [154] for segmentation. The global thresholding Otsu method is used for segmentation. Global thresholding methods partition images into regions based on their intensity values using the image’s grey-level histogram. For simple background removal, we select a suitable threshold T . If the intensity of the image at a point (x, y) exceeds T , then that part of the image is considered part of the foreground; otherwise, it is considered part of the background. If applied as a spatial-intensity transform, a binary mask is obtained where the intensity value of the foreground is set to 1, and the background is set to 0. Global thresholding performs poorly when there is low contrast between the background and foreground, large amounts of noise, and when the background intensity varies significantly across the image [155]. More accurate segmentation can be achieved using adaptive thresholding [156]. Another option is learning-based segmentation, but it requires substantial amounts of training data to achieve reliable segmentation [157]. However, global thresholding is a simple, easy to implement, computationally efficient [94, 156]. For this reason, we make the trade-off of poorer segmentation accuracy and robustness for computational efficiency. Moreover, global thresholding methods are applied in many studies in the literature for image-based plant phenotyping applications [147].

The Otsu method automatically selects a threshold value that maximises the interclass variance of the intensity levels between the background and foreground in the image’s grey-level histogram. This method is widely used in the literature [147, 158], and can be utilised for background removal in plant image processing pipelines [92]. However, the Otsu method is sensitive to noise and lacks robustness to uneven illumination [147]. Despite the method’s shortcomings, we have selected it because Otsu segmentation yields good segmentation results, is easy to implement, and is highly efficient [158, 147].

Concretely, for each leaf image, our segmentation phase aims to segment the leaf from its background. During feature extraction, we wish to obtain textural features of the leaf using information represented in the leaf’s grayscale image. Therefore, background removal is crucial in ensuring that feature extraction considers the textural information of the leaf, rather than the background. The aforementioned Otsu segmentation method is applied to the grayscale leaf image obtained from the preprocessing phase. The result of the segmentation is a binary mask, and background removal is achieved by applying the binary mask to the preprocessed grayscale leaf image.

3.2.3 Feature Extraction Phase

After segmentation, we have the isolated leaf image in the grey-scale colour space, and proceed to feature extraction. We are only extracting textural features from metrics calculated using a grey-level co-occurrence matrix [124]. Textural features are empirically some of the best predictors of leaf disease detection [125, 126], since the occurrence of a plant disease induces specific changes in the texture of the leaf [22]. Thus, we choose textural features because we can construct feature vectors from these texture metrics to train a classifier to detect the presence or absence of a tomato leaf disease [159, 22, 126].

Gray-level co-occurrence matrices (GLCM) [124] measure the statistical spatial distribution of gray-level tones, and several parameters can be calculated from these matrices to investigate the textural properties of an object in an image [160]. GLCM derived textural features are widely used in the literature to detect plant leaf diseases [126, 22].

Following the approach in [22], we compute a GLCM from the grayscale leaf image produced by the segmentation phase. Thereafter, we extract the following parameters from the GLCM: energy, contrast, sum of squares, correlation, entropy, sum entropy, cluster shade, cluster prominence, and homogeneity metrics [126, 124, 22]. Together, all these calculated GLCM parameters form a feature vector, which we will use in our classification mechanism, as discussed in the next section.

3.3 Classification Mechanism

For our classification mechanism, we use a support vector machine (SVM) [20] to determine if a tomato leaf is diseased or healthy. SVM is a supervised machine learning method that has been shown to be an effective classifier of leaf diseases [22, 1, 161, 152]. SVM classifies several different classes by finding an optimal hyperplane that linearly separates them. SVMs can learn non-simple datasets and generalize well [1].

To obtain the training dataset, we apply our aforementioned image processing steps to obtain texture feature vectors. The feature vector is labeled as healthy if it is generated from a healthy PlantVillage tomato leaf image; otherwise, it is labeled as diseased.

With this training dataset, we train our Support Vector Machine using the radial basis function kernel [162]. To avoid overfitting and bias from the dataset, we perform 10-fold cross-validation [163]. In this training methodology, the data set is divided equally or nearly equally into 10 subsets. Then, cross-validation is performed 10 times, where one set is selected as the test set, and the union of the others forms the training set. The cross-validation procedure is repeated 10 times.

3.4 Analysis

We aim to measure the reliability of our binary SVM tomato classifier by evaluating its F-measure, accuracy, precision, recall, actual positive rate, and true negative rate [163]. These metrics are commonly used to measure the performance of binary classifiers and predictive models [164]; therefore, their use allows comparison of our system with other studies. Additionally, we measure the average classification time of the system, which is the time it takes for classification to occur from the start of preprocessing to the end of classification.

Precision Suppose there are N samples, and a classifier C predicts that a sample x is positive and it is positive, then this prediction is called a true positive. If the classifier predicts x is positive when it is not, then this prediction is called a false positive. We denote TP as the total number of true positives, and FP as the total number of false positives. Then precision is the ratio of all true positives over all positives observed. Symbolically,

$$Precision = \frac{TP}{TP + FP}. \quad (3.1)$$

Accuracy Using the terminology defined above accuracy is the ratio of all correct predictions over the total number of samples. Note that analogously the TN is a prediction for which a classifier correctly predicts a sample is negative.

And FN is the converse that is if the sample is predicted as negative when it is not. Then symbolically accuracy is defined as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}. \quad (3.2)$$

Recall Also known as true positive rate or a classifiers ability to correctly identify true positive samples is defined as:

$$Recall = \frac{TP}{TP + FN}. \quad (3.3)$$

F1-score F1-score is the harmonic mean or weighted average of precision and recall. F1-score takes false positive and false negative predictions into account. Symbolically:

$$F1 = 2 * \frac{Precision * Recall}{Precision + Recall}. \quad (3.4)$$

Results & Discussion

4.1 Software

4.1.1 PlantCV

For our experiment’s preprocessing, We used PlantCV, an open-source Python library built on OpenCV and Matplotlib, because PlantCV offers modular functions for common plant phenotyping workflows. The abstractions provided by this library allows us to quickly implement and evaluate our image processing pipelines, and classification mechanisms. Furthermore, the paper’s code is extensible, and our results are more reproducible because PlantCV is open-source. Also, many contributors continuously optimize for modern hardware, so the use of this open-source software is more economical and performant than developing custom solutions from scratch.

4.1.2 Scikit-image

A forked and modified version of the open-source scikit-image library was used for feature extraction. This library is used to calculate the gray-level co-occurrence matrix (GLCM) from a grayscale image. However, the original implementation in this library did not allow the background to be excluded from its calculation, therefore, then texture from leaves and their background would be considered. Practically, this means that our feature vectors would not truly represent features of leaves, and therefore possibly lead to poor leaf disease detection performance.

4.1.3 Scikit-learn

The scikit-learn library was used to train, calculate, and evaluate metrics for the proposed SVM model.

4.1.4 Study Code

The code for this research can be found at the following GitHub repository <https://github.com/Darshan-Pillay/Honours-Research-Project>.

4.2 Hardware

Model training and preprocessing were performed on a 14-inch MacBook Pro (2021) running macOS Sonoma 14.7.2, with an M1 Pro chip and 16 GB of RAM.

4.3 Dataset

4.3.1 Plant Village

The dataset used in this study is PlantVillage. It was previously available from the official PlantVillage website; however it is no longer accessible. So, we obtained a copy of the original dataset from <https://github.com/spMohanty/PlantVillage-Dataset>. This dataset contains images of various plant species' leaves, including potato, pepper and tomato. For our experiment, we used a subset of PlantVillage, which includes 16,011 images comprising both healthy and diseased tomato leaves. These images were preprocessed to generate feature vectors for training our SVM model.

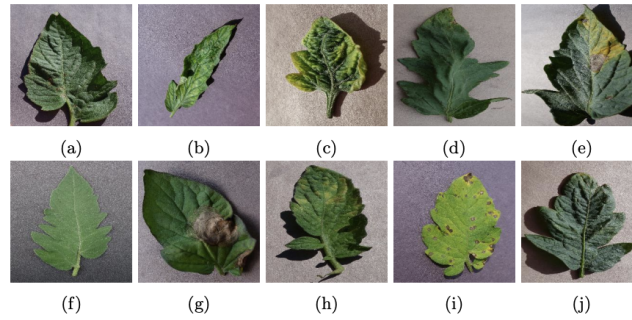


Figure 4.1: A visualization of all tomato leaf image types in the PlantVillage dataset. The following types are contained in the dataset: Target spot (a), Mosaic virus (b), Yellow leaf curl virus (c), Bacterial spot (d), Early blight (e), Healthy (f), Late blight (g), Leaf mold (h), Septoria leaf spot (i), and Spider mites (j).

4.4 Preprocessing

Each training image was preprocessed using grayscale conversion, histogram equalization, median filtering, Otsu thresholding, and background subtraction. Preprocessing enhanced the images to improve the effectiveness and accuracy of subsequent feature extraction and model training.

4.4.1 Grayscale conversion

The original RGB images were converted to grayscale. We need to do this step because our features are derived from Haralick texture statistics, specifically the GLCM, which requires grayscale input. Diseased leaves have surface texture variations, while healthy leaves are mostly uniform, making texture18 based features appropriate for leaf disease detection. See 4.2 which compares the original RGB image to its grayscale form

the original RGB image to its grayscale form

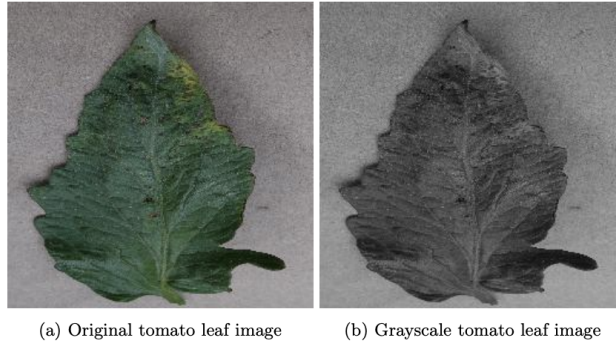


Figure 4.2: RGB to grayscale image comparison

4.4.2 Histogram equalization

Leaf images in the Plant Village dataset are captured under a variety of lighting conditions, which complicates segmentation. In bright light, the background can become illuminated and may appear to be a part of the leaf, making global thresholding methods ineffective. In our experiment, we selected Otsu segmentation for its simplicity and computational efficiency. This method of segmentation separates the foreground from the background by maximizing interclass variance, and works best when the image histogram is bimodal. However, uneven lighting typically results in unimodal or irregular histograms, so segmentation becomes unreliable. By equalizing the histogram of the grayscale image, the image's intensity distribution becomes more uniform and closer to bimodal, which enhances contrast between the foreground and background as shown in

figure 4.3. Effectively, histogram equalization improves the performance of subsequent segmentation.

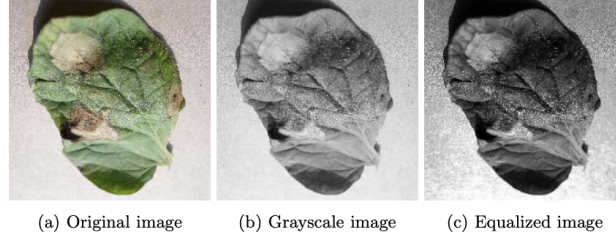


Figure 4.3: Side by side comparison of images showing the contrast enhancement effect of our histogram equalization step.

4.4.3 Median Filtering

Median filtering is applied to the equalized grayscale image to reduce noise. Noise can introduce sharp intensity changes and these abrupt variations can cause parts of the image to be incorrectly classified as part of background rather than part of the leaf when using global thresholding methods. This will reduce the overall effectiveness of our segmentation, so we apply median filtering.

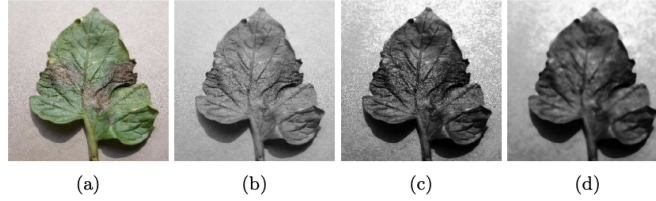


Figure 4.4: Application of median filtering in our experiment. (a) original image, (b) grayscale image, (c) histogram equalized grayscale image, and (d) median filtered equalized grayscale image.

4.4.4 Segmentation

In our segmentation step, we apply Otsu thresholding to the median-filtered image to produce a binary mask. Using this binary mask, background subtraction is then performed on the grayscale image to segment the leaf from the background as figure 4.5 shows. Our implementation assumes that the background and foreground have sufficient contrast. That is the median-filtered image has a bimodal or near-bimodal histogram distribution. Although the PlantVillage

dataset contains images captured under varying conditions, the lighting was generally uniform, making this assumption reasonable. However, this segmentation method may not generalize well to field environments or poorly lit conditions. Despite this, Otsu's method is simple to implement and computationally efficient, since one of the goals of our research is developing a cost-effective solution for low-cost greenhouse environments, this trade-off in field-applicability for economicity is acceptable.

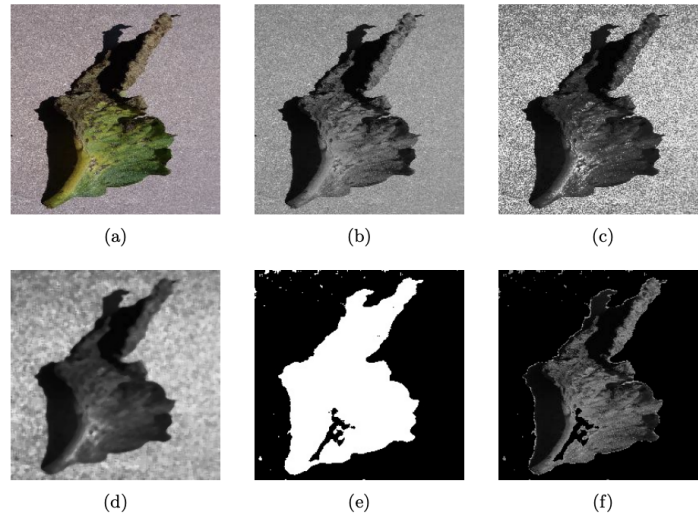


Figure 4.5: A leaf image going through all phases of preprocessing, and concluding with segmentation: (a) original image, (b) grayscale image, (c) histogram equalized grayscale image, (d) median filtered equalized grayscale image, (e) Otsu binary mark, and (f) segmented image.

4.4.5 Feature Extraction

After image segmentation, the GLCM matrix is computed from the segmented grayscale image, considering only the gray levels corresponding to the leaf. Should we consider all gray levels, then the quality of our features will be affected and consequently our classifier’s performance will suffer. Therefore, the accuracy of the segmentation directly influences the quality of the extracted features. So, we assume that all regions excluded from the segmented image are non-plant areas. That is gray levels of value 0 are ignored. Without this assumption, the GLCM is a matrix $P(i, j, \theta, d)$, representing the probability that gray levels i and j occur at an angle θ and distance d away from each other. But with 0 gray levels ignored, then the GLCM represents $P(i, j, \theta, d|partofleaf)$. That is the GLCM gives us leaf textural information and not background texture metrics.

The GLCM matrix requires a choice of distance d and angle θ for its computation. Higher values of d are more computationally efficient, but give less spatial textural information. Since we do not use any color information for features, high d values may lead to less accurate predictions as our model will be trained with lower textural resolution data. So, we select 1 as our value for d . As for the angles, we computed the matrix for 0° , 45° , 90° and 135° . By calculating the GLCM for multiple angles we get textural information in many different directions. These 4 angles are sufficient since computing the GLCM for angles like 180° , 225° , 270° and 315° is essentially just the transpose of GLCMs for 0° , 45° , 90° and 135° .

For each GLCM obtained from a distance d and angle θ we calculate several harlick features from it. Namely as mentioned in the methodology section these metrics are: energy, contrast, homogeneity, correlation, entropy, angular second moment, dissimilarity, mean, variance, standard deviation, and homogeneity. Since, there are 10 metrics and 4 angles, we compute 40 features for each image. Each feature vector is labelled as healthy or diseased depending on if it was obtained from a healthy leaf image or a diseased leaf image.

4.4.6 Dataset Distribution & Split

The PlantVillage dataset is very imbalanced and biased towards images of diseased leaves. Less than 10% of the tomato leaf images are labelled healthy. This means that a model trained on this data, will be biased towards images of diseased leaves and struggle to differentiate between healthy and non-healthy leaf images. Therefore, the chance of false positives may be greater. In the context of precision agriculture where the minimization of pesticide application is desired, a biased model like this could be costly and undesirable should agriculture decisions be based off the models predictions. Therefore, in choosing dataset split for training and test data, we ensured that the overall distributions of diseased and healthy leaves remained the same, so that our classifier would

have greater chances of generalizing well to unseen data. Specifically we used 90% of the PlantVillage tomato leaf data subset as training data, and 10% we held out for testing. For model evaluation, we opted for stratified 10-fold cross evaluation since the dataset is imbalanced. Essentially, the split is stratified to preserve the class distributions and mitigate bias to prevent overfitting.

4.4.7 SVM Classifier Training

We have selected an SVM as our classification mechanism. That is we have trained a binary classifier that takes in as input harlick texture feature vectors from the GLCM of a segmented leaf image, and outputs whether or not the feature belongs to a healthy leaf or diseased tomato leaf.

We did not perform an extensive search and comparison to select our hyperparameters, but instead used the radial basis kernel for the SVM which, as mentioned in the literature review, has performed well and has shown good generalization capabilities in leaf disease detection studies. Additionally, we use the weighted SVM variant since our dataset is imbalanced so that healthy leaf images are given more consideration since they are in the minority. This helps to reduce the risk of overfitting. Once the hyperparameters were fixed, we then trained and had the model fit the training data.

4.5 Model Evaluation

The results of our experiment are presented below. The tables in 4.2 and 4.1 show the performance of our model on the training and data set. We can that for the training and test data that our model achieved 83% and 84% accuracy respectively. However this not a good measure of the overall effectiveness and performance of our model. In fact accuracy on its own is a flawed measure. If the model were only trained on 99% of leaf images, and 1% of healthy leaves then it could predict that all images are of diseased leaves and achieve 99% accuracy. But as it is exposed to more unseen data it may not generalize well and its accuracy in real terms would be much poorer. Therefore, we also examine the precision, recall and f1-score of our classifier.

Taking a look at the tables we see that recall is 84%. Recasting recall in probabilistic terms, it measures the probability that our classifier will classify a feature vector as a diseased if the feature is from a diseased leaf image. Let us denote this as $P(T|D)$ where T is the event that the classifier classifies a feature as diseased, and D is the event that the feature is really from a diseased leaf image. It is then the case that $P(T|D) = 1 - P(F|D)$, so the probability that our classifier gives a false negative is only 16% on the training data, so our models capacity to rule out tomato leaf diseases is good. Our model performed similarly on the test data. The relative portions of false negatives and true positives in the confusion matrices 4.6 and 4.7 also confirm this claim.

As for precision our model achieved 97% on the training data, and 97% on the test data. This suggests our model even on unseen data is able to predict that a leaf image has a disease. On the other hand our dataset was imbalanced, so this metric is not very good on its own. However, the f1 scores for on the training data and test data is 90%. This high value is because recall and precision are quite close to one another, suggesting our classifier has a good balance of low false negatives and accurate predictions. This is further supported by the ROC and AUC digrams in 4.9 and 4.8.

The ROC curve in 4.8 shows that our our model on the training data strikes a good balance between the recall and the false positive rate. The ROC measures an implicit variable, in the sense that if $P(DiseasedImage)$ is the probability that an image is for a diseased leaf image, if we pick a bound B and classify any image as diseased if $P(DiseasedImage) > B$, then we get a False positive rate and true positive rate (Recall) which are plotted as points, and form the ROC. Taking many different values of B we get the resulting curve. Overall, the curve deviates from the identify ROC through the origin meaning it is more likely to give meaningful predictions. The ROC curve 4.9 is similar to the training data ROC, and so this suggests our SVM classifier generalizes well to the unseen test data. Furthermore, the AUC (area under the curve) in both figures is close to 1 at least 0.90 for both. An AUC of 1 is what a perfect classifier would achieve, so our classifier in terms of this metric performs well in detecting tomato leaf diseases.

Dataset size	Accuracy	Precision	Recall	F1
14409 images	83%	97%	84%	90%

Table 4.1: Summary of our SVM classifier’s average performance on the training data running after 10-fold stratified cross validation.

Dataset size	Accuracy	Precision	Recall	F1
1602 images	84%	97%	84%	90%

Table 4.2: Summary of our SVM classifier’s performance on the the test data.

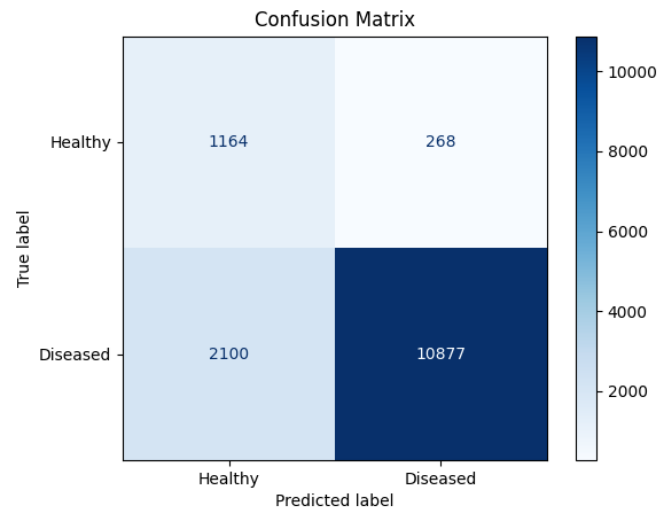


Figure 4.6: Confusion matrix for our model's prediction on the training data

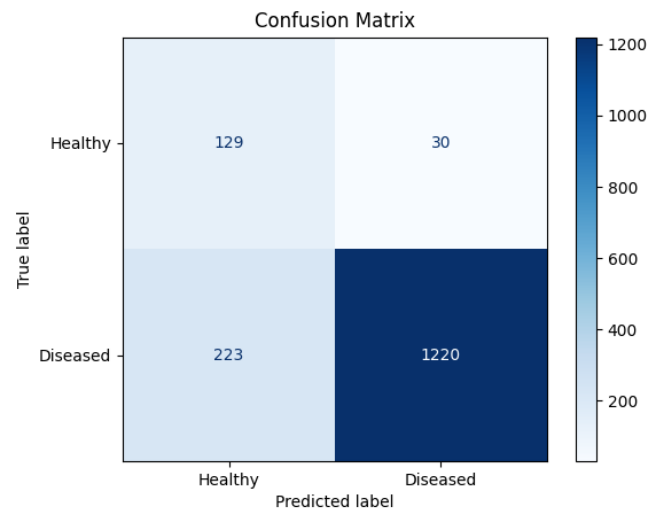


Figure 4.7: Confusion matrix for our model's prediction on the test data

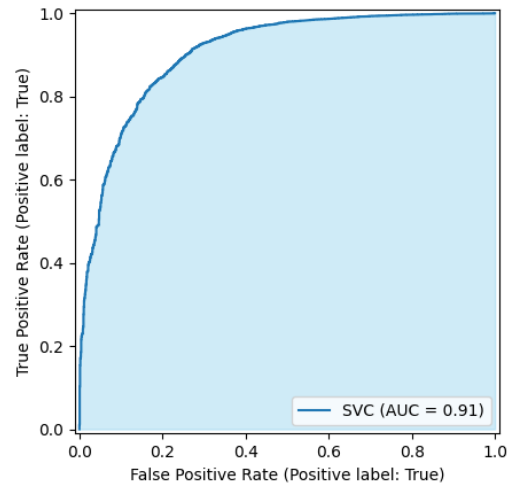


Figure 4.8: ROC and AUC for our model on the training data after 10 fold stratified cross validation.

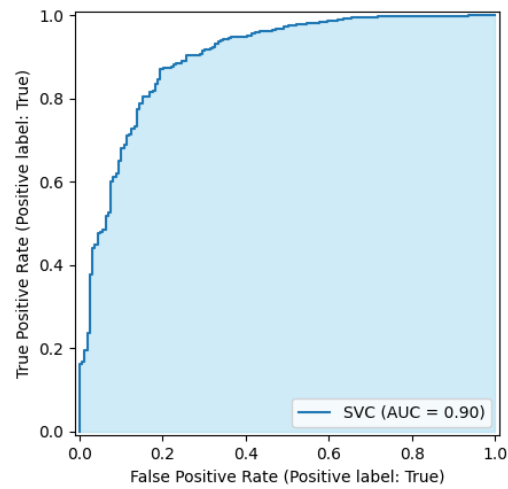


Figure 4.9: ROC and AUC for our model after predictions on the test data.

4.6 Limitations

As mentioned earlier, the PlantVillage dataset is imbalanced, and only contains less than 10% of healthy tomato leaf images. The confusion matrix 4.7 shows that our classifier struggles to differentiate between diseased and healthy leaf images. Even though, we reduced the bias by performing stratified test-training data split. This suggests to really be sure of the generalizability of our solutions, we need to evaluate the performance of classification on a balanced data set, or on data with a different distribution than PlantVillage.

Also, in developing an economical solution we made a trade-off in terms of segmentation methods. Opting for simple low complexity segmentation like Otsu thresholding is more computationally efficient, but it struggles with images taken with uneven lighting. Therefore, the overall accuracy of feature vectors and predictions may be compromised by cheaper segmentation choices. On the other hand, machine learning and deep learning segmentation methods are more robust, but may be prohibitive in resource constrained contexts.

Additionally, textural feature vectors computed from the GLCM can be computationally expensive depending on the size of the image, and the number of gray levels. In our research we used 256x256 8-bit images, so on modern hardware these calculations are not particularly painful, but in practice these images may be substantially larger and then other considerations like quantization or binning of gray level values may need to be done to practically use our solution

Conclusion

This research set out to build an accessible open-source tool for detecting tomato leaf diseases in low-cost greenhouse settings. By leveraging modular computer vision libraries and large public datasets like PlantVillage, we developed an economical SVM classifier that achieved strong accuracy and F1 scores on its training and test data. This work extends existing literature by showing the applicability, generalizability, and effectiveness of SVM classifiers for tomato leaf disease detection. Despite challenges with dataset imbalance and the limits of Otsu segmentation in poor lighting conditions, our solution remains practical for greenhouse use and can be easily verified, extended, or modified. Most importantly, this project shows that efficient and adaptable disease detection is possible with open-source tools. Reproducible, shareable, and modular approaches mean plant phenotyping does not have to be reserved for those with large capital or proprietary software. In future, this method can be turned into a mobile application, making tomato leaf disease detection more accessible to growers and communities everywhere.

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